Classical statistical inference

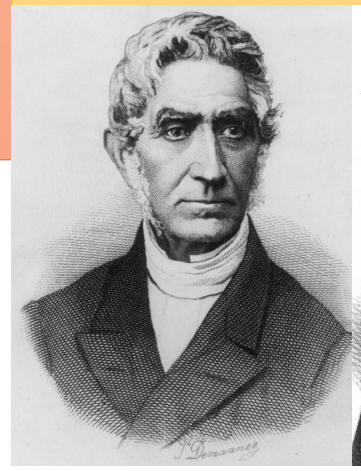
Part 2

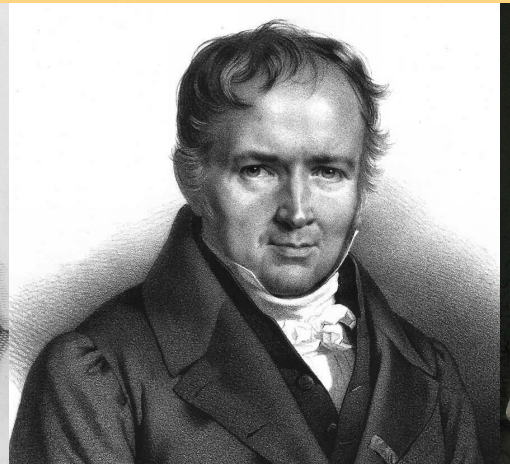
Associated notebook:

04-Basic_statistical_inference_frequentists_2/Frequentist_inference_01.ipynb

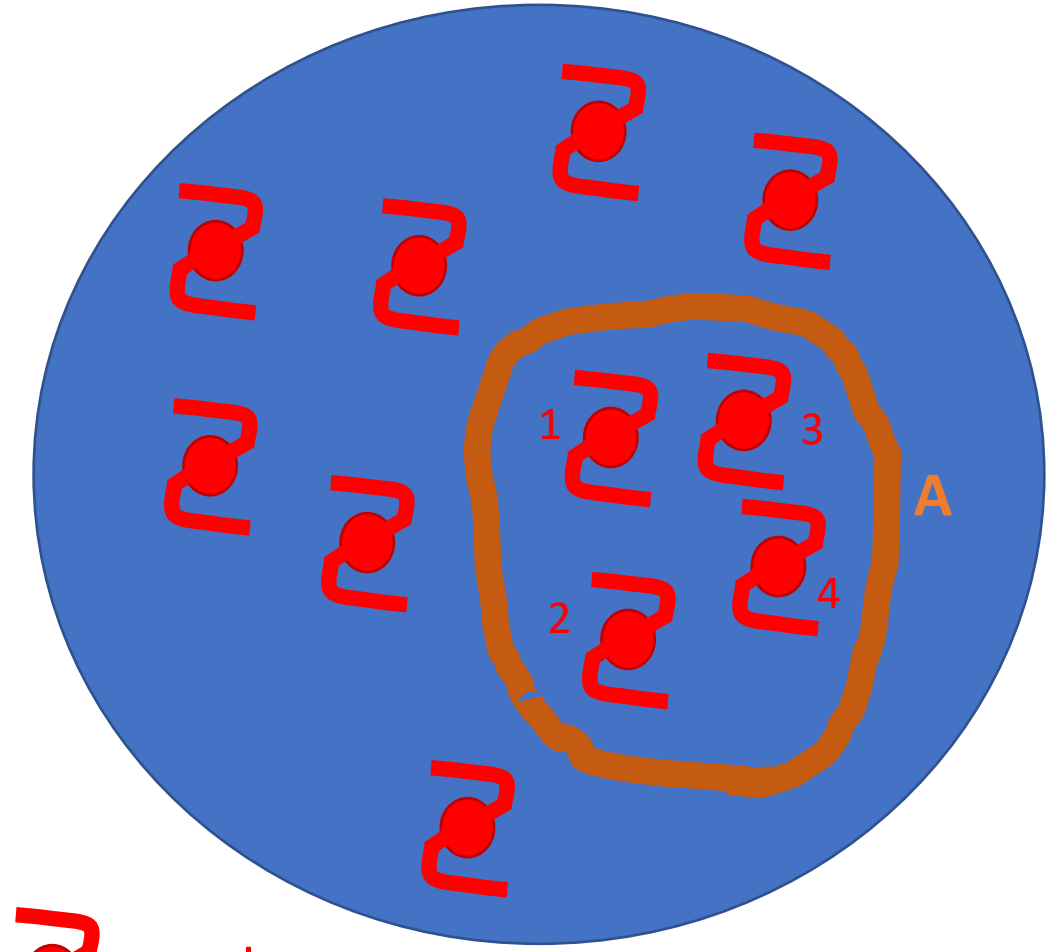
How do we do science?

Common Language	Statistics Language
1. We repeat **observations / experiments**	1. Obtain a **sample** out of a *population*
2. We make **measurements** (under a set of conditions)	2. We **estimate** a quantity / calculate ***p(quantity | params)***
3. We **compare** measurements to model / **derive a model** from the data and aim at predicting **a (range of) future measurement(s)**	3. We do **modelling**, ***hypothesis testing***, and derive **confidence intervals**





Point estimate $\hat{\theta}$



= galaxy

Example:

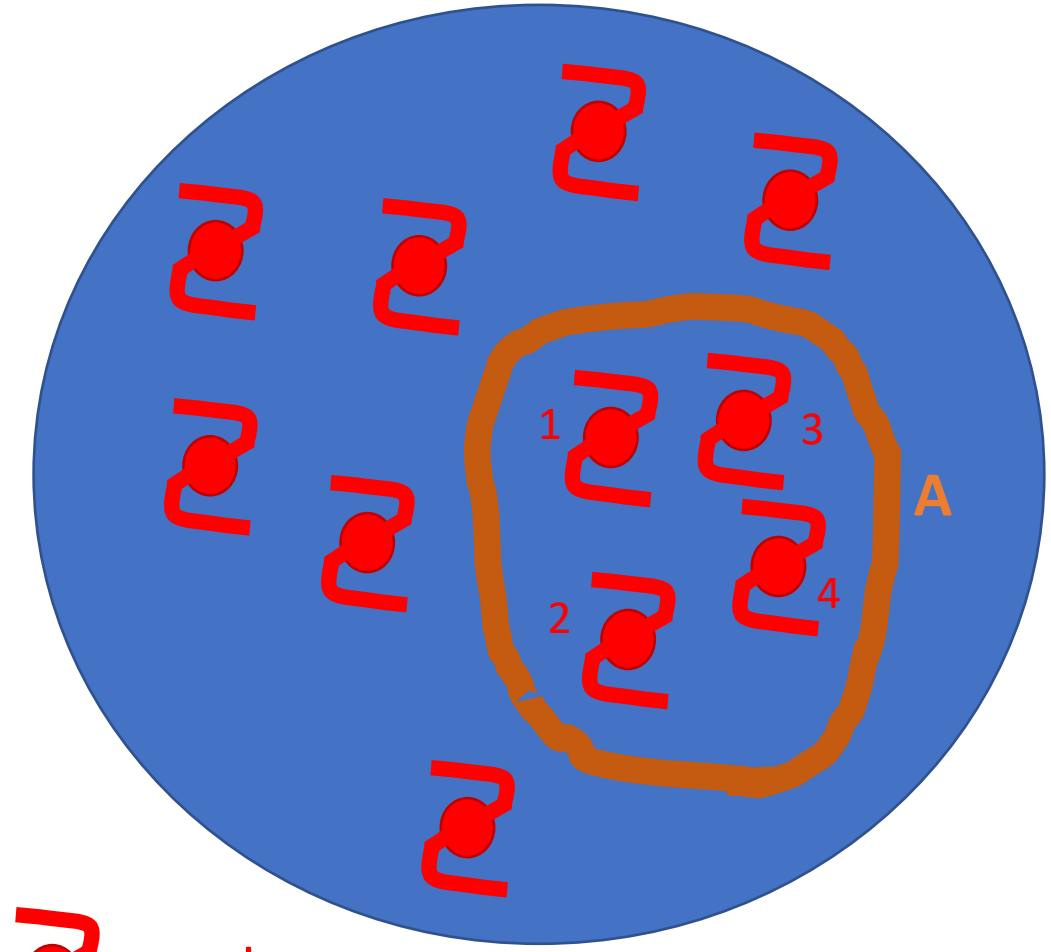
θ = mean mag. of a population of galaxies

A: Your data set = subsample of measurements:
A = $\{X_1, X_2, X_3, X_4\}$ where X = mag. (this is a RV)

$$\hat{\theta} = \frac{X_1 + X_2 + X_3 + X_4}{4}$$ ≡ Point estimate of θ

If you do the experiment with another sample (⇒ different *realisation*) you will get another $\hat{\theta}$

Point estimate $\hat{\theta}$



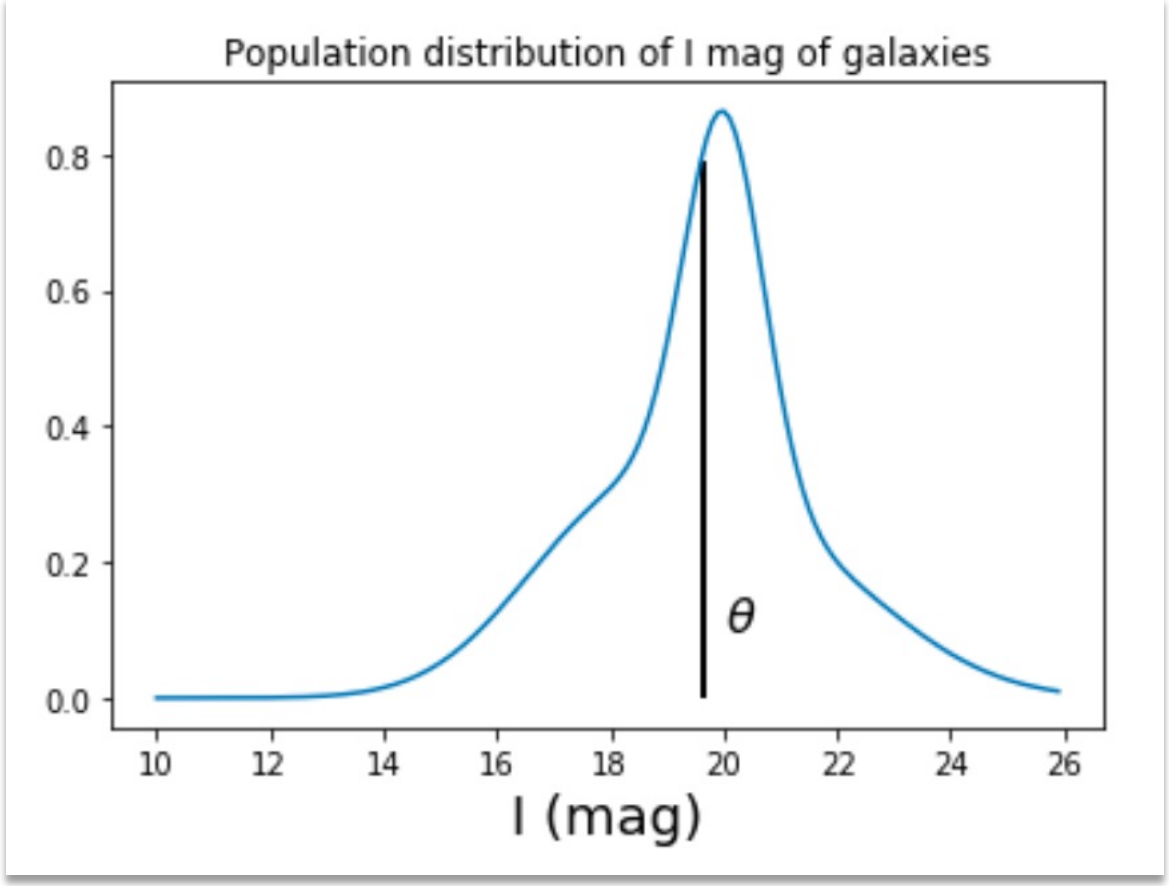
= galaxy

Generalisation:

$$\hat{\theta} = g(X_1, X_2, X_3, \ldots X_n)$$

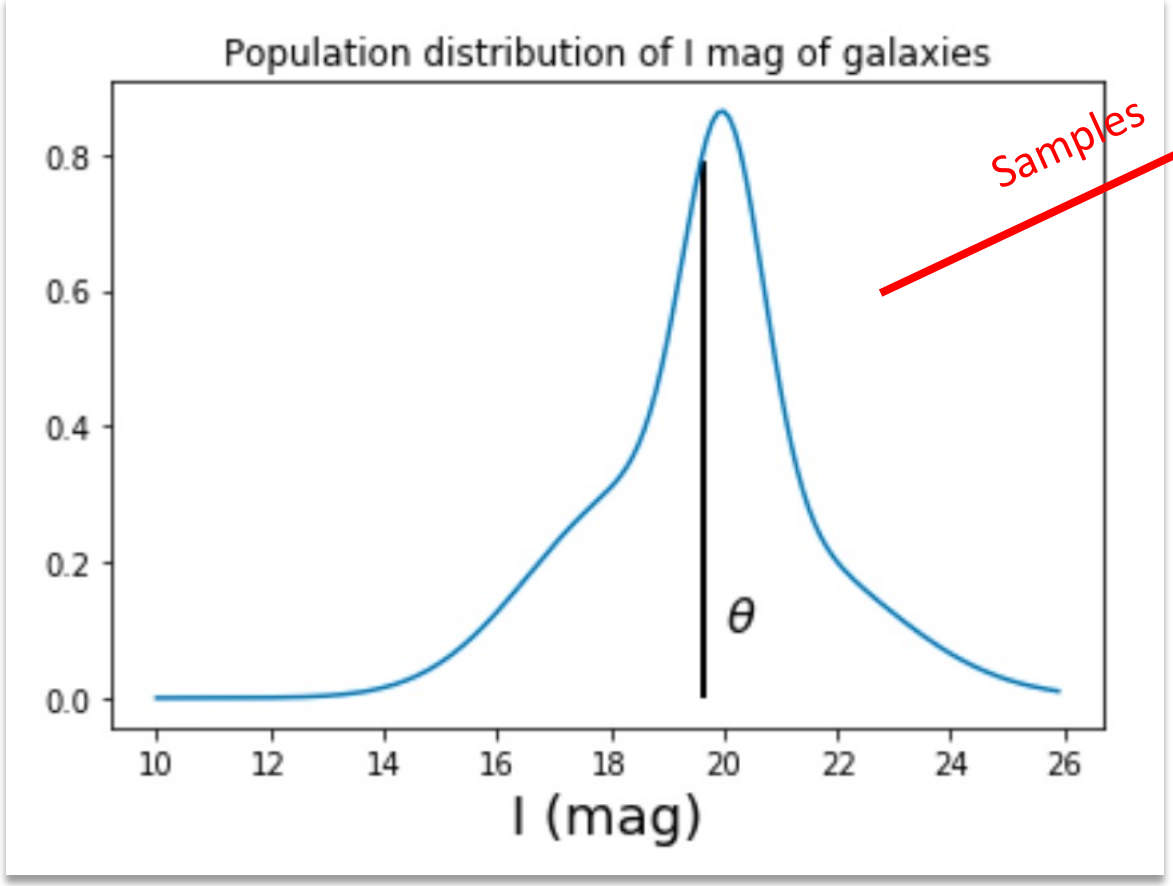
- Point estimate of a param. is a *function* of RV X_1, …
- It is as well a **Random Variable (RV)**
- It can be biased, is characterized by a variance but should ideally be consistent (converges towards θ)
- Distribution of $\hat{\theta}$ is called *sampling distribution*

Point estimate $\hat{\theta}$



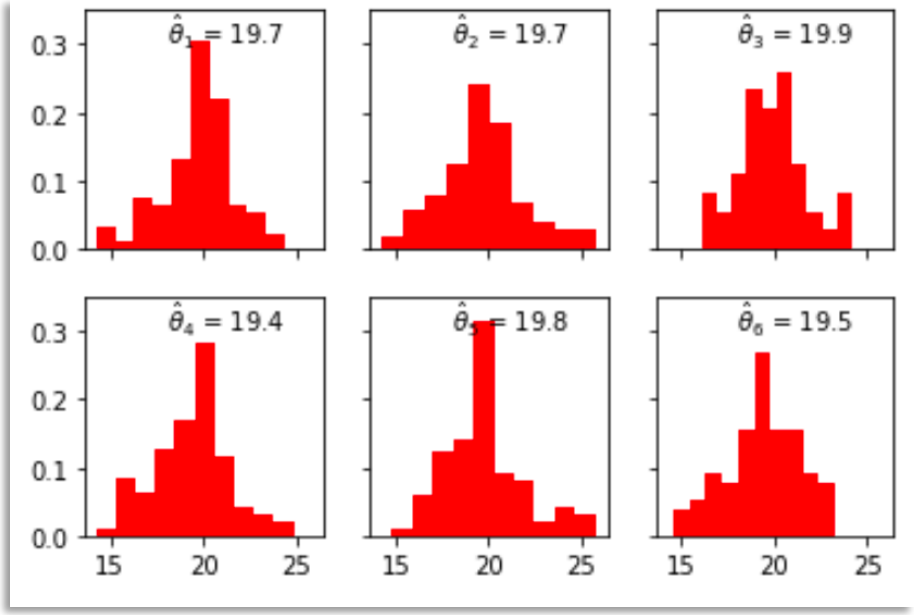
Population mean: $\theta = 19.66$

Point estimate $\hat{\theta}$



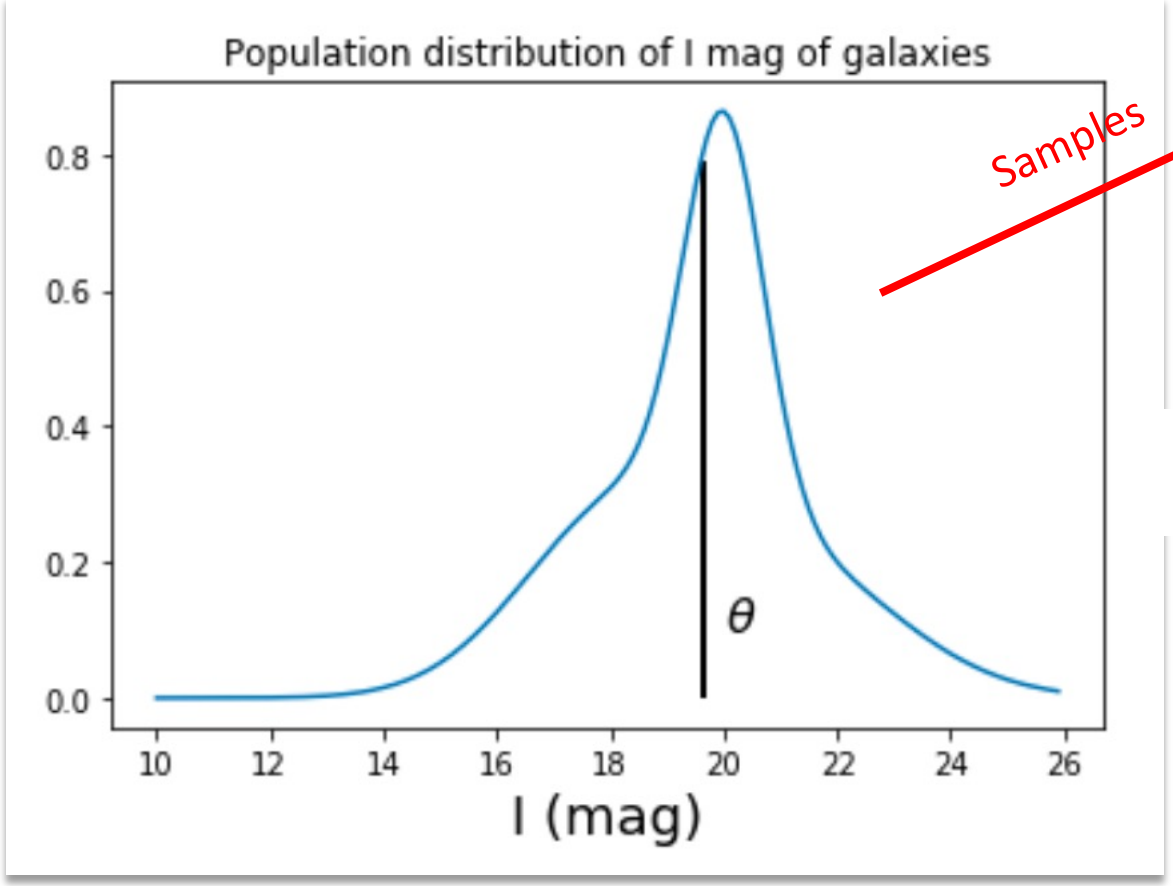
Samples

Six different *samples* drawn from the true population



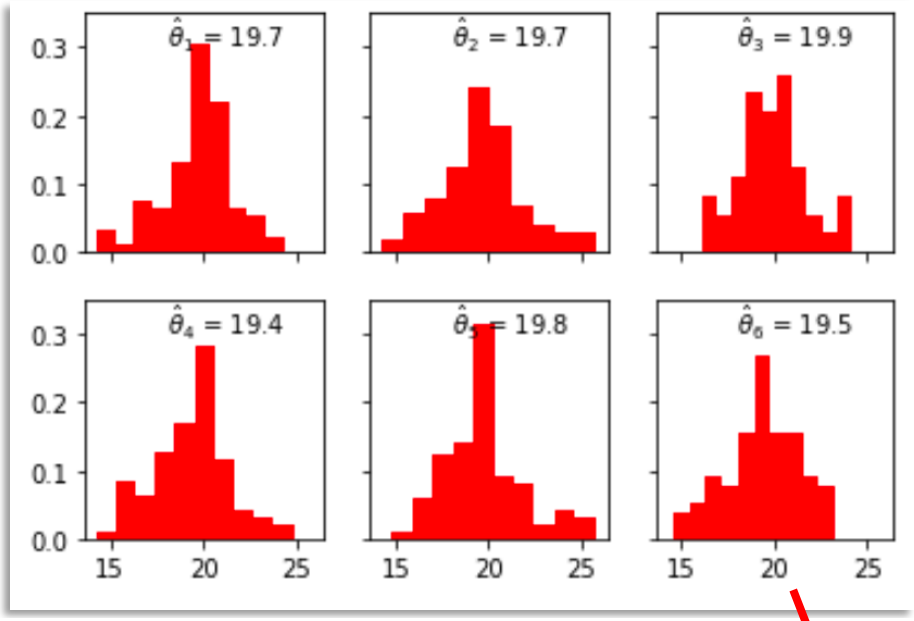
Population mean: $\theta = 19.66$

Point estimate $\hat{\theta}$

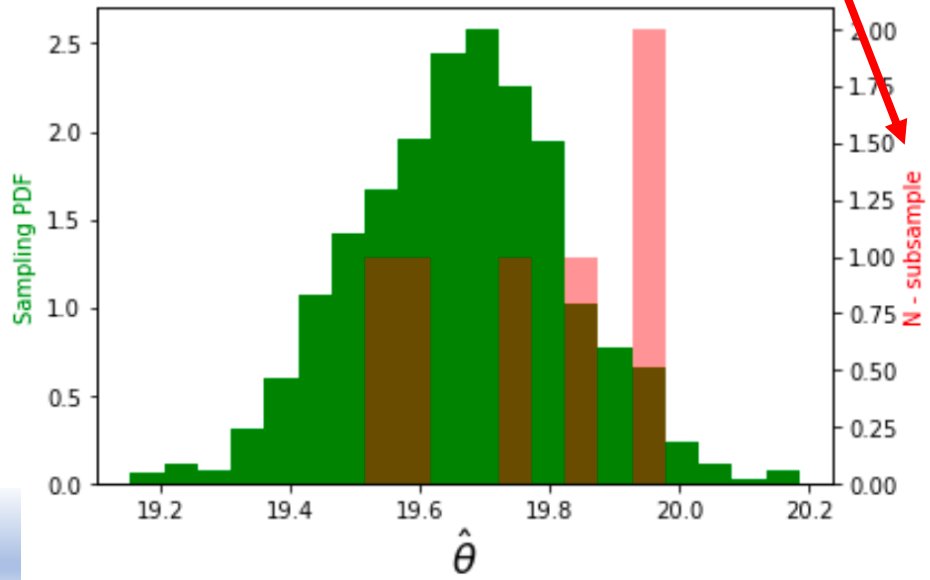


Samples

Six different *samples* drawn from the true population



Distribution of $\{\hat{\theta}_1, \hat{\theta}_2, \ldots, \hat{\theta}_k\}$ = sample distribution



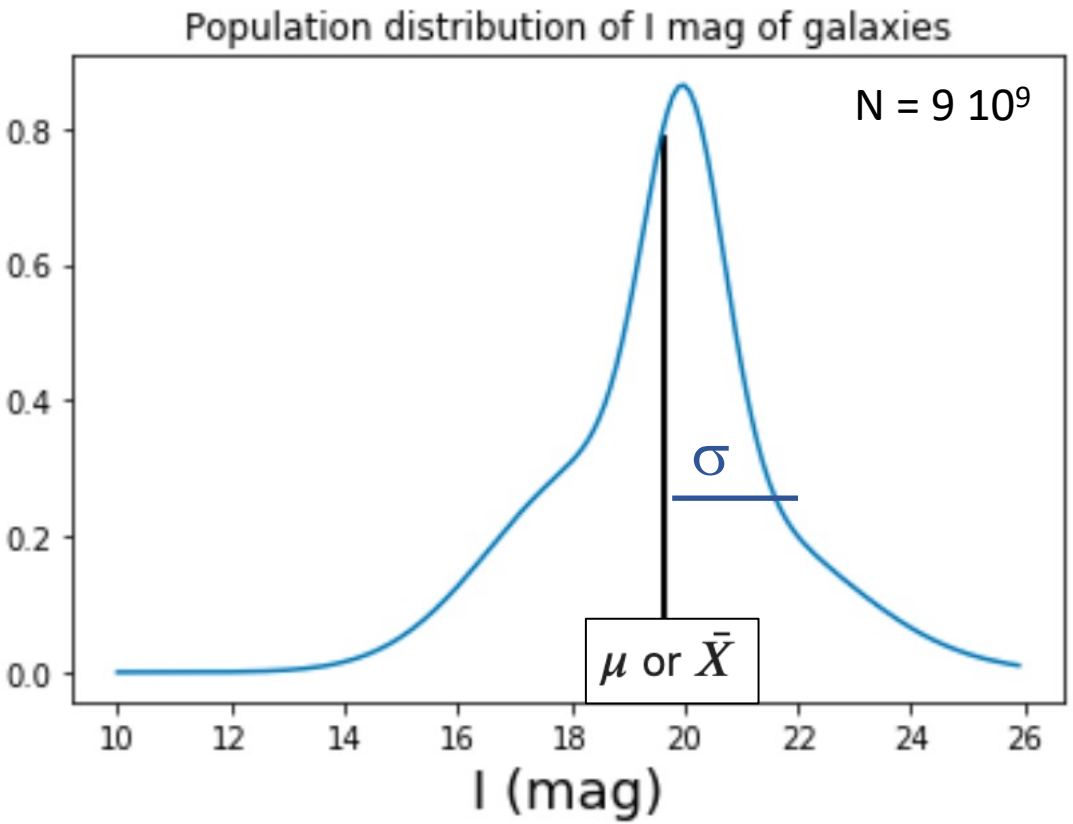
Population mean: $\theta = 19.66$

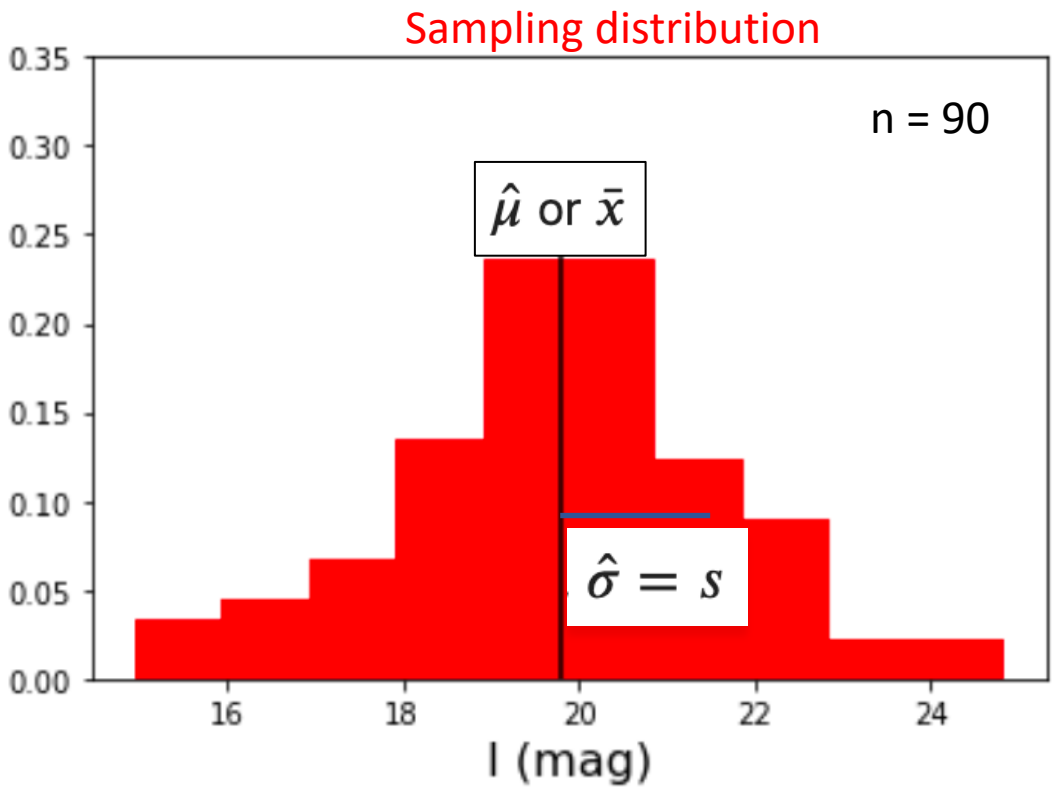
Go to: Sect. II.1 of the notebook

Summary statistics

Name	Population Statistics	Sample Statistics
size	N	n
mean	$\mu = \bar{X} = \frac{\sum_i X_i}{N}$	$\hat{\mu} = \bar{x} = \frac{\sum_i x_i}{n}$
Variance	$\sigma^2 = \frac{\sum_i (X_i - \bar{X})^2}{N}$	$s^2 = \frac{\sum_i (x_i - \bar{x})^2}{n-1}$
Standard deviation	$\sigma = \sqrt{\sigma^2}$	$\hat{\sigma} = s = \sqrt{s^2}$

Bessel's correction





Central limit theorem

When independent **random variables** are added, their *sum* tends towards a normal distribution (if n >>)

- This is true even if the original RV are not normally distributed
- Sampling dist. of mean tends (for large *n*) towards a Normal distribution
- Sampling dist. of variance (for large *n)* does NOT tend towards a Normal distribution

Go to: Sect. II.1.1. of the notebook

Summary (sample) statistics: *standard error*

Standard error (stde) ≠ Standard deviation (std)

Name	Formula
Standard error on the mean	$stde(\bar{x}) = \frac{s}{\sqrt{n}}$
Standard error on the sample stdev	$stde(s) = s/\sqrt{2(n-1)}$
Standard error on proportions	$stde(\hat{p}) = \frac{\sqrt{p\,(1-p)}}{\sqrt{n}}$
Standard error on p^{th} percentile (h_p = PDF at p^{th} percentile)	$stde(q^p) = \frac{1}{h_p}\frac{\sqrt{p\,(1-p)}}{\sqrt{n}}$

Go to: Sect. II.2.1. of the notebook

Distribution of sample quantities: **Not** always Normal distribution

If RV $\{X_i\}$ whose population is distributed as $N(\mu, \sigma)$

- Sample distribution of $\hat{\mu} \sim N(\mu, \sigma/\sqrt{n}) \iff Z = \dfrac{\mu - \hat{\mu}}{(\sigma/\sqrt{n})} \sim N(0, 1)$
- Sample distribution of $t = \dfrac{\mu - \hat{\mu}}{(s/\sqrt{n})} \sim t(n-1)$ s is derived from the sample dist.

  Student distribution
- Sample distribution of $S = \dfrac{(n-1)s^2}{\sigma^2} \sim \chi^2(n-1).$

  Chi square distribution

Estimator efficiency and robustness

- The median of a distribution ($\tilde{x}$ or q^{50}) is a more robust estimator of the "*location*" μ of a distribution but there are 2 drawbacks:
  - It is slower to calculate the median
  - The median is about 25% less efficient than the mean (for a Gaussian distribution). It means that you need a 25% larger sample to reach the same accuracy on the mean.
- The normalised interquartile range (IQR = $q_{75}-q_{25}$) is a more robust estimator of the scale parameter than *s*. For a gaussian distribution $s_G = k * (q_{75}-q_{25})$ with k = 0.7413. You can also calculate the Median Absolute Deviation: $\mathrm{MAD} = \mathrm{median}(|\mathrm{x} - \tilde{\mathrm{x}}|)$

  One has $s_G = k * \mathrm{IQR} = 2\mathrm{k} * \mathrm{MAD}$

See Sect. II.3.1. and II.3.2. of the notebook